

# TrialReady LK

AI-Assisted Driving School ERP & Statutory Regulatory Compliance Platform

Technology Challenge Competition (TCC) Module (CCS2360 / CCS3361) • BSc (Hons) in Cyber Security

GROUP 03

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## 2. SYSTEM OVERVIEW

**TrialReady LK** is a specialized enterprise cloud ERP engineered to solve severe operational inefficiencies and statutory compliance risks faced by Sri Lankan driving schools. The application automates mandatory student milestones under the *Department of Motor Traffic (DMT)* and *National Transport Medical Institute (NTMI)*, prevents training schedule collisions, and uses machine learning to predict practical trial success.

### Target Users

- Academy Admins:** Centralized multi-branch control, fleet tracking, audit records, and billing.
- Instructors:** Daily session roster, attendance logging, and student maneuver scoring.
- Learners (Students):** 7-stage DMT tracker, computerized theory drills, and readiness status.

### Key Platform Features

- 7-Stage Pipeline:** Tracks medicals, 3-month permit maturation, and trial dates.
- Conflict-Free Scheduling:** Prevents dual-booking of vehicles and instructors.
- DMT Compliance Slips:** 1-click generation of official A4 Student Logbooks & Trial Slips.

### Core AI/ML Functions

- Trial Readiness Predictor:** Multi-factor probability scoring (0–100%) and tiering.
- Adaptive Remedial Quiz:** Generates targeted drills from mock exam weak areas.
- Intelligent Copilot:** Instant natural language operational assistant.

## 3. SYSTEM REQUIREMENTS & ACCESS

### Environment & Software Requirements

- Operating System:** Windows 10/11, macOS 12+, Linux, Android 11+, or iOS 15+.
- Web Browsers:** Google Chrome (v110+), Microsoft Edge, Firefox, or Safari.
- Network:** Standard internet connection (≥ 2 Mbps broadband or 4G/5G).
- Application URL:** `http://localhost:5173/` (Local) or Hosted Domain.

### Verified Demo Login Credentials

The sign-in interface includes **1-Click Quick Demo Login** buttons for each role:

| Role       | Email Address               | Password       | Landing Route      |
|------------|-----------------------------|----------------|--------------------|
| ADMIN      | admin@drivingschool.lk      | Admin@123      | /dashboard         |
| INSTRUCTOR | instructor@drivingschool.lk | Instructor@123 | /instructor/portal |
| STUDENT    | student@drivingschool.lk    | Student@123    | /student/portal    |

## 4. HOW TO USE THE SYSTEM (STEP-BY-STEP INSTRUCTIONS)

### 4.1 User Authentication & Portal Access

1. Open your browser and navigate to the application URL ( <http://localhost:5173/> ).
2. Enter your registered email and password, or select a role badge (**Admin**, **Instructor**, or **Student**) for 1-click demo entry.
3. Click **Sign In to Academy**. The system authenticates user roles and routes you to your authorized portal.
4. To switch accounts or terminate the session, click your profile icon on the top right bar and click **Sign Out**.

### 4.2 Student Registration & 7-Stage Learner Journey Tracking (Admin)

1. Click **Students** from the sidebar menu, then click + **Add Student**.
2. Enter the student's legal name, NIC/Passport number, phone, email, training branch, and licence class (e.g., *Class B — Dual-Purpose Car*). Click **Save Enrolment**.
3. Navigate to **Learner Journey** to track statutory compliance across all 7 stages:

**Stage 1:** Enrolment → **Stage 2:** NTMI Medical → **Stage 3:** DMT Permit (3-Month Countdown) → **Stage 4:** Theory Exam Cleared → **Stage 5:** 20h Practical Lessons → **Stage 6:** AI Readiness Approved → **Stage 7:** DMT Trial Passed.

4. To register medical or permit papers, click the student row, click **Update Medical** or **Update Permit**, enter the certificate/permit number, issue date, and expiry date, then click **Save Record**.

### 4.3 Practical Training Scheduling & Attendance Marking (Admin & Instructor)

1. From the sidebar, select **Practical Sessions**. Switch between **Calendar View** and **List View**.
2. To book a new lesson, click + **Schedule Lesson**:
  - Select the **Student**, **Licence Category**, **Instructor**, **Training Vehicle**, **Date**, and **Time Slot**.
  - *Real-Time Conflict Prevention*: The system verifies resource availability immediately. If the vehicle or instructor is already booked for that exact hour, a warning alert displays, blocking accidental double-bookings.
  - Select maneuvers to practice (e.g., *Clutch Control*, *Hill Start*, *Parallel Parking*, *3-Point Turn*), then click **Confirm Booking**.
3. To record attendance and evaluate the student after a drive:
  - Click **Mark Attendance** on the session card.
  - Set status (**Present**, **Late**, or **Absent**).
  - Check off skills mastered during the lesson, award an **Instructor Rating (1 to 5 Stars)**, enter lesson feedback notes, and click **Save Evaluation**.

### 4.4 Trilingual DMT Theory Mock Exam Simulator & Road Signs (Student & Admin)

1. Navigate to **Theory Hub** from the sidebar.
2. Select examination language using the trilingual selector: **English**, **සිංහල (Sinhala)**, or **தமிழ் (Tamil)**.
3. Click **Start Mock Exam**. The computerized exam launches with an official 45-minute countdown clock and 40 DMT questions.
4. Select answers (A, B, C, D) and navigate questions via the numbered navigator palette.
5. Click **Finish & Submit Test**. The system provides an instant score report:
  - Score percentage, correct/incorrect count, and official pass determination (DMT threshold: ≥ 75%).
  - Click **Review All Questions & Explanations** to view Highway Code rationales and vector road sign illustrations for every mistake.
6. Select **Road Signs Flashcards** to practice interactive flashcards across Regulatory, Warning, and Priority traffic signs.

### 4.5 Financial Billing, Packages & Payment Receipts (Admin)

1. Select **Financials** from the sidebar to inspect revenue metrics, outstanding balances, and tuition collections.
2. To assign a training course to a student, click **Packages**, choose a package (e.g., *Standard Class B — LKR 48,000*), and assign it to the student profile.
3. To record an installment, click **Record Payment**, choose the student, enter the amount paid (e.g., *LKR 20,000*), payment method (Cash/Card/Bank Transfer), and click **Save Payment**.
4. Click **View Receipt** to view or print the official branded payment receipt showing remaining balance.

4.6 AI/ML PRACTICAL TRIAL READINESS EVALUATION (CORE AI WORKFLOW)

The machine learning predictive engine synthesizes five heterogeneous data dimensions into an objective trial success probability (0–100%) to eliminate premature practical trial failures at the DMT Werahera testing grounds.

TRIALREADY LK — END-TO-END PREDICTIVE AI WORKFLOW

1. User Input

20h Road Lessons, NTMI Medical, Permit Maturation (≥3 mo), Theory Tests, Maneuver Ratings

→

2. Multi-Factor ML Processing

Weighted scoring algorithm evaluates statutory criteria & risk factors (Hill start, S-bend)

→

3. Generated Output

Score (0–100%), Tier Classification Badge, and Dynamic Risk Checklist

→

4. Actionable Decision

Score ≥ 80%: Print DMT Trial Slip & Logbook.  
Score < 80%: Trigger AI Remedial Quiz.

How to Execute & Interpret the AI Predictor

1. Navigate to **Trial Readiness** in the sidebar.

2. Click on any student to open the **AI Trial Predictor Card**.

3. The system compiles all session ratings and compliance dates automatically.

4. Interpret the Readiness Tier Badge:

- TRIAL READY (≥ 80%)** : Fully qualified for official DMT test.
- NEARLY READY (65–79%)** : 1–2 polishing sessions recommended.
- NEEDS PRACTICE (40–64%)** : Needs additional road hours.
- NOT READY (< 40%)** : Missing statutory prerequisites.

Executing Adaptive AI Remedial Quizzes

1. On the student readiness page, inspect the **Deficient Risk Factors** (e.g., *Regulatory Signs, Roundabout Priority*).

2. Click **Launch AI Remedial Quiz**.

3. The AI engine generates a targeted 5-to-10 question drill isolating the candidate's exact identified weaknesses.

4. Upon completion, the candidate's readiness score recalibrates dynamically in real time.

5. If ready, click **Print DMT Admission Slip** to produce the official DMT Werahera exam pass.

5. KEY SCREEN LAYOUTS & USER INTERFACE REFERENCE

| Screen / Component                  | UI Layout & Contents                                                                | User Purpose & Primary Action                                    |
|-------------------------------------|-------------------------------------------------------------------------------------|------------------------------------------------------------------|
| Fig 1: Login & Role Selector        | Multi-branch picker, credentials form, and 3 1-click demo account badges.           | Quick authentication into Admin, Instructor, or Student portals. |
| Fig 2: Executive Admin Dashboard    | Real-time KPI cards (Active Students, Fleet Utilization, Revenue, Today's Lessons). | Central command overview for academy directors and owners.       |
| Fig 3: Learner Journey Pipeline     | 7-stage progress stepper, permit countdown badges, and statutory upload modals.     | Monitors student legal compliance and milestone progression.     |
| Fig 4: AI Trial Readiness Predictor | Circular score gauge (0–100%), readiness tier pill, and risk factor checklist.      | Evaluates student trial readiness and triggers remedial quizzes. |
| Fig 5: Theory Mock Exam Hub         | 40-question exam view, 45-min timer, trilingual toggle, and review explanations.    | Simulates DMT computerized theory test with road sign visuals.   |
| Fig 6: Practical Session Scheduler  | Weekly calendar grid, instructor conflict alert banner, and attendance modal.       | Schedules driving drives and marks maneuver ratings (1–5 stars). |

6. ERROR HANDLING & TROUBLESHOOTING GUIDE

| Issue Encountered                      | Potential Root Cause                                                                           | Recommended User Solution                                                                                                                      |
|----------------------------------------|------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| Invalid Login / Access Denied          | Incorrect password or unauthorized user role.                                                  | Verify credentials or click the <b>1-Click Quick Demo Login</b> button (Admin / Instructor / Student) on the sign-in screen.                   |
| Schedule Conflict Alert (Amber)        | The selected instructor or dual-control vehicle is already scheduled for that time slot.       | The system prevents double-booking. Choose an open time slot, assign another available vehicle, or select a different instructor.              |
| Trial Booking Locked / Ineligible      | Student has < 20 practical hours, expired medical, or learner permit has not reached 3 months. | Navigate to <b>Learner Journey</b> . Complete required driving sessions, update the NTMI certificate, or wait for statutory permit maturation. |
| Exam Interruption / Accidental Refresh | Network drop or accidental tab close during computerized mock exam.                            | TrialReady LK automatically caches test progress in local storage. Return to <code>/theory/exam</code> and click <b>Resume Test</b> .          |

TrialReady LK — TCC User Manual

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